STA 445 S26 Assignment 5

2026-03-17

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

sept abbreviation isnt correct

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs. wont work because it doesnt use a three letter abbreviation like sep.

e. 07-Dec-1941.

```
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
mdy("1-5-1998")
```

```
## [1] "1998-01-05"
```

could be wrong because 1 could be the day where 5 would then be the month

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1998")
```

```
## [1] "1998-05-21"
```

know its correct because there is not a month 21 so it must be the day

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am", tz = "US/Pacific")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-05-05 10:20am", tz="America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:20:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
dob <- mdy("Oct 6, 2004")
today <- today()
```

a. Calculate the date of your 64th birthday.

```r
dob + years(64)
```

```
## [1] "2068-10-06"
```

b. Calculate your current age (in years).

```r
year(as.period(dob %--% today))
```

```
## [1] 21
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
bday <- dob + years(year(as.period(dob %--% today)) + 1)
bday
```

```
## [1] "2026-10-06"
```

d. The number of *days* until your next birthday.

```r
as.period(today %--% bday, unit = "days")
```

```
## [1] "203d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(today %--% bday)
```

```
## [1] "6m 19d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
(phon_call <- ymd_hm("2015-05-08 15:00", tz = "US/Arizona") %>% with_tz(call, tzone = "Pacific/Au
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

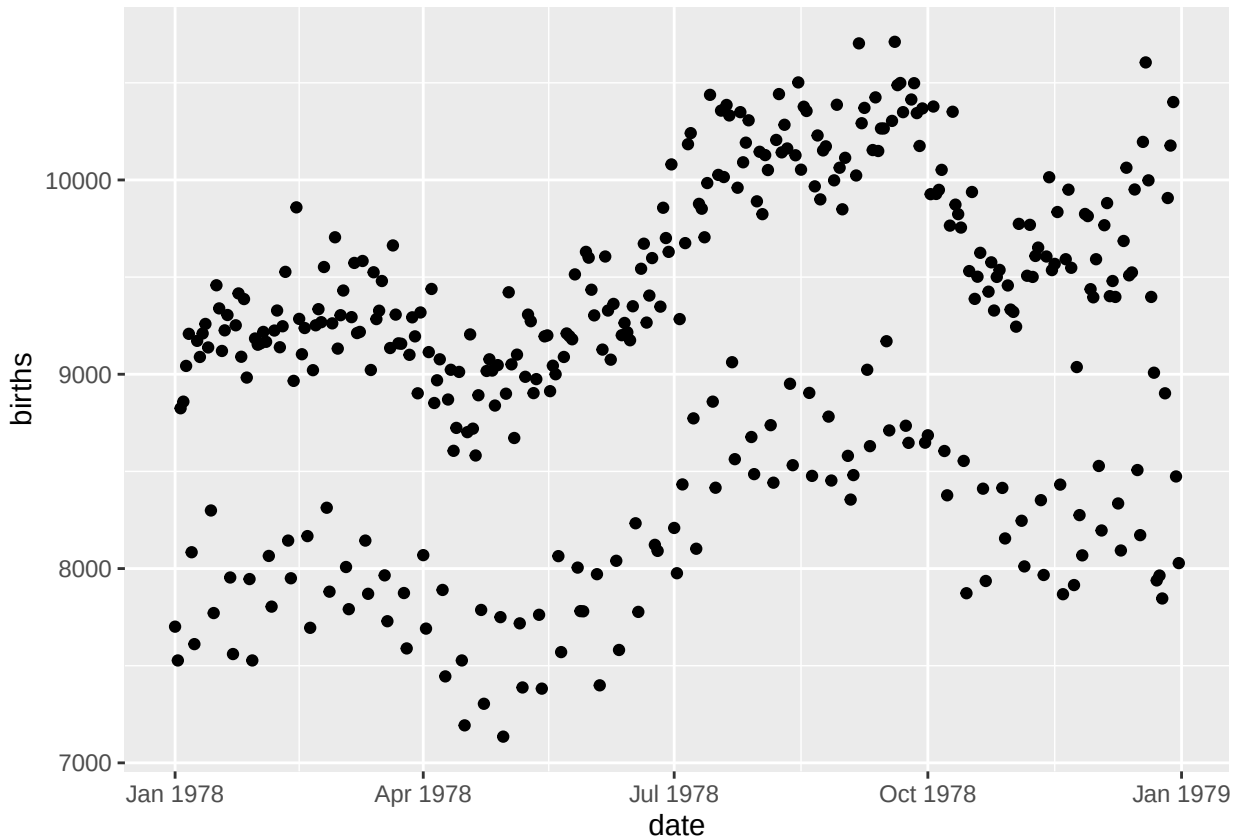
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
data("Births78")
births <- Births78 %>% select(date, births)
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
ggplot(births, aes(x = date, y = births)) + geom_point()
```



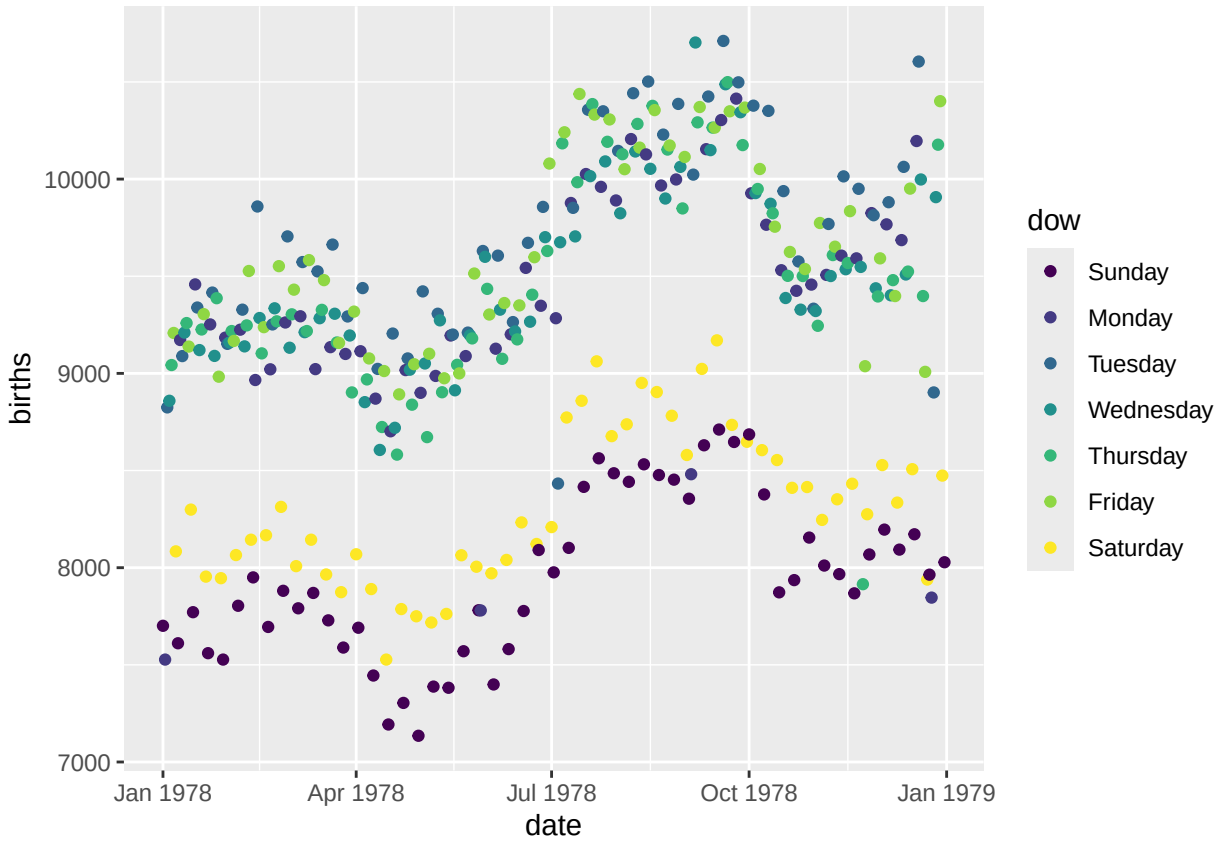
the graph repeats a pattern where it drops then raises again and continues this trend with a trend below the top one repeating the same pattern but averaging less births, i think this trend could be caused from there being less births on weekends, being the bottom trend

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```r
births <- births %>% mutate(dow = wday(date, label = TRUE, abbr = FALSE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```r
ggplot(births, aes(x = date, y = births, color = dow)) + geom_point()
```


10000
9000
8000
7000
births
Jan 1978
Apr 1978
Jul 1978
Oct 1978
Jan 1979
date
dow
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday